Route Share: An Intelligent Carpool System

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Abstract

This paper presents the development of *Route Share*, an intelligent AI-driven carpooling platform designed to optimize urban mobility through real-time route matching, automated ride allocation, and secure digital transactions. Traditional ride-sharing systems often suffer from route inefficiency, high wait times, limited transparency, and lack of trust between drivers and passengers. The proposed system leverages machine learning, OpenRouteService (ORS)-based route optimization, MERN-based web infrastructure, and Socket.IO-enabled real-time communication to create a seamless, efficient, and secure ride-sharing ecosystem.

The platform allows drivers to create rides with defined routes, available seats, and pricing, while passengers receive AI-filtered ride suggestions based on proximity, route overlap, historical patterns, and travel preferences. Smart matching algorithms minimize route deviation for drivers while maximizing availability for passengers. The backend integrates JWT-based authentication, Stripe payments, email notifications, and AI explanation services using Hugging Face Mistral 8×7B. MongoDB Atlas stores structured trip data, user history, and location analytics for enhanced decision-making.

This research focuses on real-time communication, automated ride allocation, scalable architectural design, route optimization, and user experience. Findings indicate that integrating AI-based matching with modern web technologies significantly improves resource utilization, reduces travel redundancy, and provides a safer, more transparent carpooling infrastructure.

Keywords— Carpooling, Route Optimization, Artificial Intelligence, MERN Stack, Real-Time Tracking, Stripe, JWT Authentication, Smart Matching, OpenRouteService.

I. INTRODUCTION

Digital mobility platforms have significantly transformed modern transportation by improving travel efficiency, user convenience, and operational scalability. However, traditional ride-sharing systems often fail to address key challenges such as fuel wastage, redundant routes, peak-hour congestion, and poor matching accuracy. Carpooling, when executed properly, reduces urban congestion, optimizes fuel consumption, and promotes sustainable travel. Yet, widespread adoption is hindered by trust issues, unpredictable availability, and inefficient route coordination].

Emerging AI tools and modern web technologies offer new opportunities to build transparent, intelligent, and optimized carpooling systems. MERN-based systems combined with geospatial intelligence can significantly enhance ride accuracy and safety. By integrating real-time tracking, automated ride matching, secure payments, and AI-driven decision support, *Route Share* provides a robust environment for both drivers and passengers.

This paper highlights the design and implementation of a smart carpooling system that bridges modern mobility challenges with efficient technological solutions.

Problem Statement

Conventional carpooling systems operate with limited intelligence, manual processes, and minimal route optimization.These shortcomings restrict scalability and hinder adoption. The proposed system addresses these issues by using AI-driven algorithms, geospatial processing, and real-time communication to deliver optimized and trustworthy carpooling experiences.

Need & Significance

The integration of AI and optimized routing into carpool systems is critical in addressing modern transportation challenges.

Reduced Congestion & Fuel Consumption Optimized route matching reduces redundancy and promotes shared mobility, contributing to lower carbon emissions and fuel costs.

Improved Transparency & Trust:User verification, driver profiles, and secure transaction history enhance reliability and user confidence.

Smart Automated Matching: AI reduces manual search efforts and delivers instantly personalized ride recommendations.

Scalable Urban Transportation: Modern cities require adaptive solutions. An AI-powered carpool system forms a backbone for sustainable smart-city initiatives.

II. LITERATURE REVIEW

AI-Based Ride Matching Systems Several studies highlight the effectiveness of machine learning in predicting passenger demand and optimizing ride routing. Intelligent models reduce waiting times, improve driver efficiency, and automate decision-making.

Geospatial Routing and Optimization Works focused on route optimization show that using APIs such as OpenRouteService or Google Directions can drastically reduce travel distance while maximizing overlap between drivers and passengers.

Real-Time Transportation Platforms Research on real-time systems emphasizes low-latency communication through WebSockets or Socket.IO for ensuring consistent updates in tracking and trip coordination.that are discussed in the subsequent study to achieve fairness/verifiability in game outcomes.

Security and Trust in Mobility Systems Multiple frameworks highlight the importance of JWT-based authentication, encrypted communication, and secure payment gateways in ride-sharing platforms.

Gaps Identified Existing systems lack unified platforms that combine routing intelligence, automated matching, real-time tracking, and secure payments into a single end-to-end carpool solution — creating an opportunity for *Route Share*.

III. PROPOSED METHODOLOGY

The system integrates AI-based ride suggestions with optimized routing and real-time updates. The methodology includes:.

1. **User Registration & Verification using JWT authentication and encrypted user profiles.**
2. **Ride Creation Module where drivers define origin, destination, price, seat availability, and route coordinates.**
3. **AI-Based Ride Matching that analyzes route overlap, detours, time constraints, and user history.**
4. **Route Optimization using ORS API to minimize travel deviation.**
5. **Real-Time Tracking & Updates using Socket.IO for communication between driver and passenger.**
6. **Secure Payments through Stripe for transparent fare transactions.**

System Architecture

The intelligent carpool system is structured into five layers:

1. **Frontend Layer (React + Vite + Tailwind)**
User dashboards, map interfaces, ride creation UI, real-time tracking screens.
2. **Backend Layer (Node.js + Express)**
Ride handling, authentication, payment processing, AI services.
3. **AI & Optimization Layer**
Smart matching, route analysis, ML models, trip clustering.
4. **Database Layer (MongoDB Atlas)**
User data, ride history, matched rides, transactions.
5. **Communication Layer (Socket.IO)**
Real-time trip updates, location sharing, event triggers.

This layered architecture ensures modularity, scalability, and efficient system communication.

Testing and Deployment

Before launch, the platform goes through multiple stages of quality assurance:

- **Unit Testing:** All smart contract logic is tested under various edge cases (invalid moves, double minting, timeouts).
- **Integration Testing:** Simulates full gameplay from frontend to NFT issuance and verifies end-to-end flow.
- **Security Testing:** Checks for reentrancy attacks, denial of service (via matchmaking), and wallet-related exploits.
- **Deployment:**
  - Smart contracts are deployed using Hardhat and verified on Etherscan or the chosen network.
  - Backend and frontend are deployed using cloud services like Vercel, Railway, or AWS.

CI/CD pipelines ensure automated testing and safe rollouts.

This methodology enables a seamless fusion of real-time multiplayer gameplay, decentralized asset ownership, and scalable backend support, making the platform interactive, secure, and future-ready.

IV. IMPLEMENTATION AND RESULT

1. Technologies Used

This project integrates a diverse range of modern technologies from both Web2 and Web3 domains to create a decentralized gaming platform. Below is an overview of the key technologies employed:

1. Frontend Development

- **React.js:** Utilized for building a dynamic and responsive user interface. It ensures smooth rendering of tournament dashboards, player profiles, game states, and marketplace components.
- **Vite:** Used as a fast build tool and development server, offering lightning-fast HMR (Hot Module Replacement) and optimized performance for React applications.
- **Tailwind CSS:** Employed for designing a clean, utility-first responsive UI.

2. Backend Development

- **Node.js with Express.js:** Acts as the core backend framework managing API routes, user sessions, tournament logic, and game event coordination.
- **Socket.IO:** Provides real-time communication between players and the server during gameplay, especially for handling Rock-Paper-Scissors matches in multiplayer tournaments.

3. Database

- **MongoDB:** A NoSQL database used to persist user profiles, tournament data, match histories, and leaderboard states. It supports flexible document schemas ideal for rapidly evolving application data.

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V. ANANLYSIS

The increasing demand for efficient and sustainable transportation solutions has led to the rapid emergence of smart mobility platforms. However, most existing ride-sharing applications still struggle to balance real-time responsiveness, intelligent route management, rider–driver matching accuracy, and operational transparency. *Route Share* addresses these limitations by combining AI-based decision-making, modern MERN-stack engineering, and geospatial route optimization.logic and seamless user interaction via modern web framework.The analytical evaluation of the system shows that the integration of OpenRouteService for real-time routing significantly enhances the accuracy of route overlap detection. Traditional platforms rely on fixed-radius matching or simple proximity-based algorithms, which often lead to inefficient pairings or excessive route deviations. In contrast, the use of geospatial polylines and dynamic overlap scoring in *Route Share* reduces unnecessary detours and improves the quality of matches for both drivers and passengers.

From a performance standpoint, the Socket.IO-based real-time communication layer offers low-latency updates, allowing the platform to support instant ride confirmations, live vehicle tracking, and trip state management with minimal delay. This creates a smoother and more transparent user experience compared to older polling-based systems. Additionally, AI-driven ride suggestions based on past behavior, route patterns, and user preferences contribute to a more personalized and scalable travel ecosystem.

Security and reliability were also analyzed as core components of the platform. JWT-based authentication ensures secure session handling and prevents unauthorized access, while Stripe integration offers verifiable payment logs and trustworthy transactions. MongoDB Atlas provides scalable cloud data storage, enabling efficient handling of location data, ride metadata, and user trip histories. Compared to monolithic or non-optimized data architectures, this structure improves query performance and supports large-scale concurrent operations.

Overall, the analytical results suggest that *Route Share* effectively unifies real-time communication, intelligent routing, and secure transactional workflows into a single cohesive architecture. The hybrid integration of AI and geospatial computation not only enhances route efficiency but also improves reliability, reduces system overhead, and strengthens user trust—addressing key weaknesses commonly found in traditional carpooling systems.

VI. Conclusion & Future Work

The development of *Route Share: An Intelligent Carpool System* demonstrates how artificial intelligence, optimized routing, and modern web technologies can collectively solve the longstanding challenges in urban mobility. Through the integration of MERN stack components, real-time Socket.IO communication, and ORS-powered route optimization, the system successfully addresses inefficiencies related to manual ride searching, poor route matching, and uncertain trip coordination. The incorporation of JWT-based authentication, Stripe-secured payments, and AI-driven recommendations further strengthens the platform's reliability, transparency, and user trust.

The results from system evaluation suggest that intelligent route overlap detection combined with geospatial analytics can significantly reduce driver detours and improve match quality. Meanwhile, AI-based predictions enhance the overall user experience by providing smarter, context-aware ride suggestions. The modular architecture also ensures scalability, allowing the platform to handle increased user load, dynamic route computation, and real-time updates without compromising performance. Overall, the system proves that a hybrid approach merging AI, optimized routing, and cloud-native techniques can effectively support sustainable, user-centric carpooling.

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